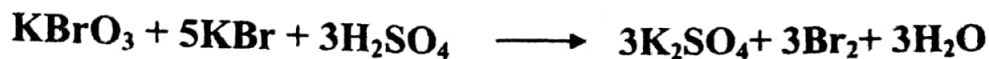


Experiment: Determination of the activation energy for the reaction,



Theory:

The relationship between the **time (t)** taken to reach a given extent of the above reaction and the **rate constant (k)** is given by,

$$\ln t = -\ln k + \text{constant}$$

The **time (t)** taken to reach a given extent of the above reaction is found adding a fixed amount of phenol and some Methyl Orange indicator to the reaction mixture. Bromine formed reacts rapidly with phenol and after the phenol has been consumed, any further Bromine formed instantaneously bleaches the indicator.

Procedure:

1. Prepare a mixture containing 10.00 ml of 0.010 M aqueous phenol, 5.00 ml of 0.50 M Sulphuric acid and 4 drops of Methyl Orange indicator in a boiling tube. Maintain the temperature at 25°C. Add 10.00 ml of Bromide/Bromate solution into the boiling tube which is maintained at the same temperature. Start your stop watch while adding nearly a half of Bromide/Bromate solution. Stir the reaction mixture well. Measure out the time (t) taken for the red colour to disappear in the reaction medium. **(0.08333 M KBr, 0.0167 M KBrO₃, i.e. equivalent to 0.050 M Bromine)**
2. Repeat (1) at 15 °C, 20°C, 30°C, 35°C.
3. Write down the relationship between **rate constant (k)** and the above **temperature (T)** and hence derive a relationship between **t** and **T**.
4. By using a suitable graphical method, determine the activation energy for the above reaction.